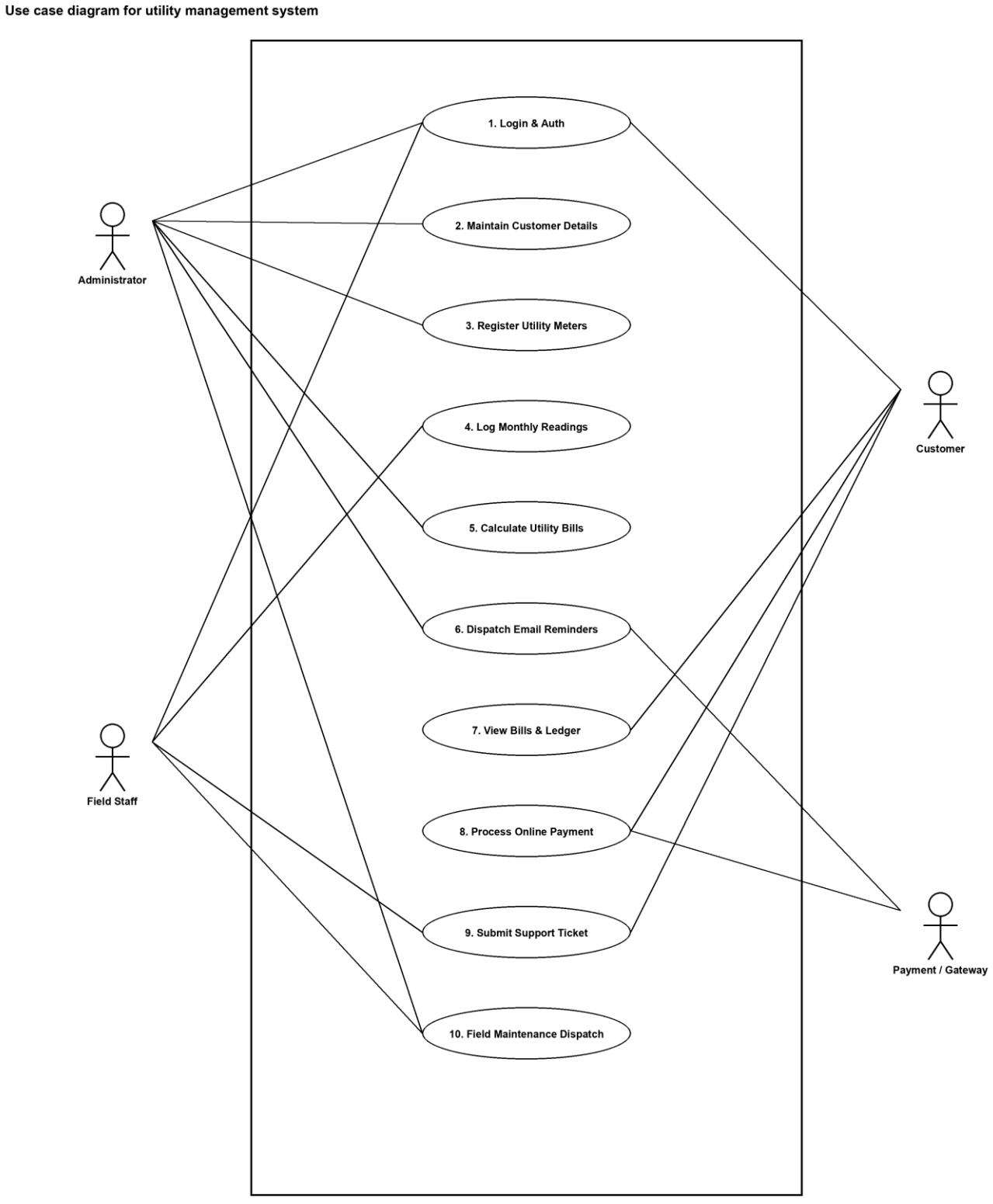
UTILITY MANAGEMENT SYSTEM

Use Case Diagram | Class Diagram | Sequence Diagrams

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Date: August 30, 2026

1. Use Case Diagram

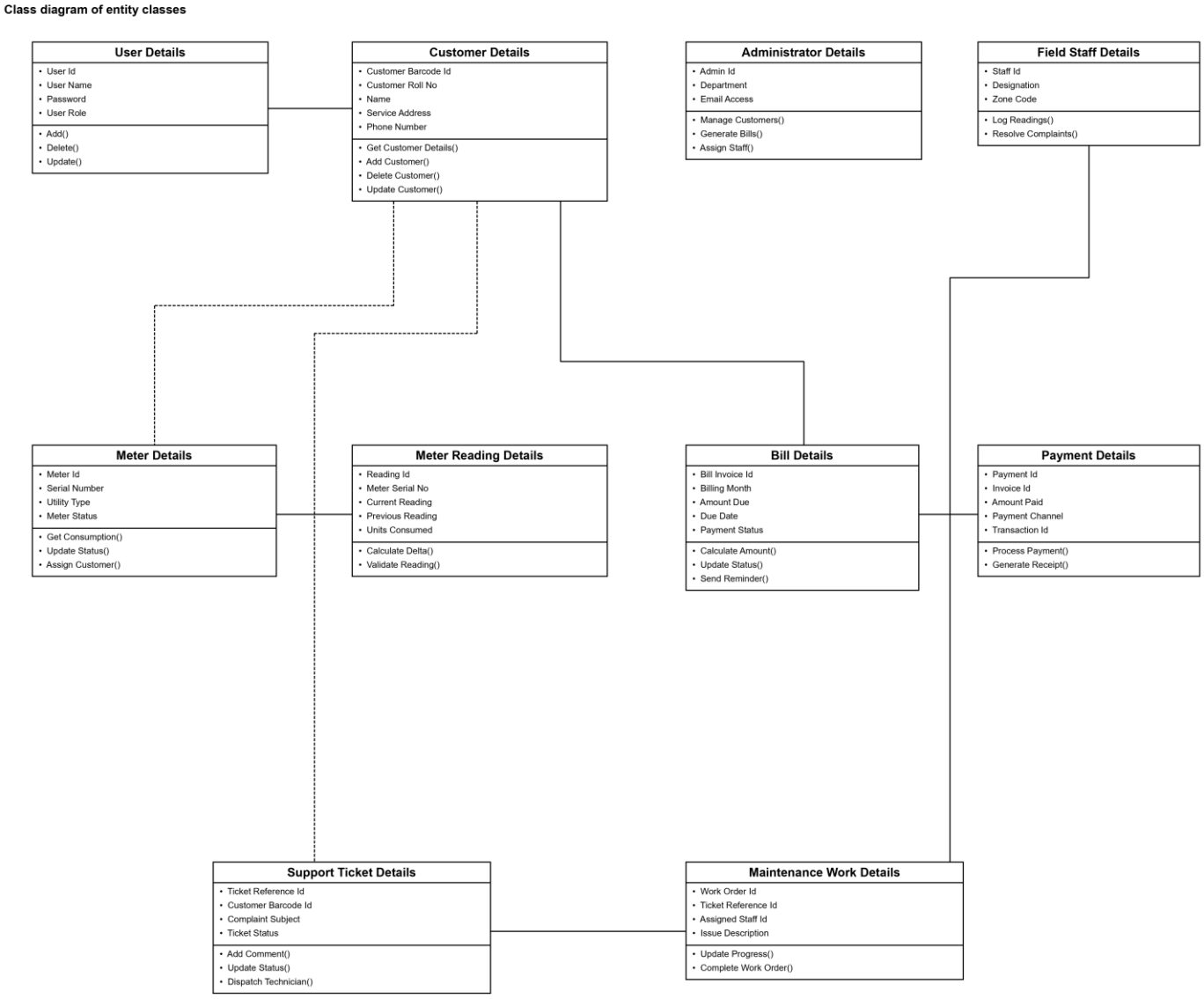
System boundary and actor interaction model for Utility Management System.



Use case diagram for utility management system

2. Class Diagram

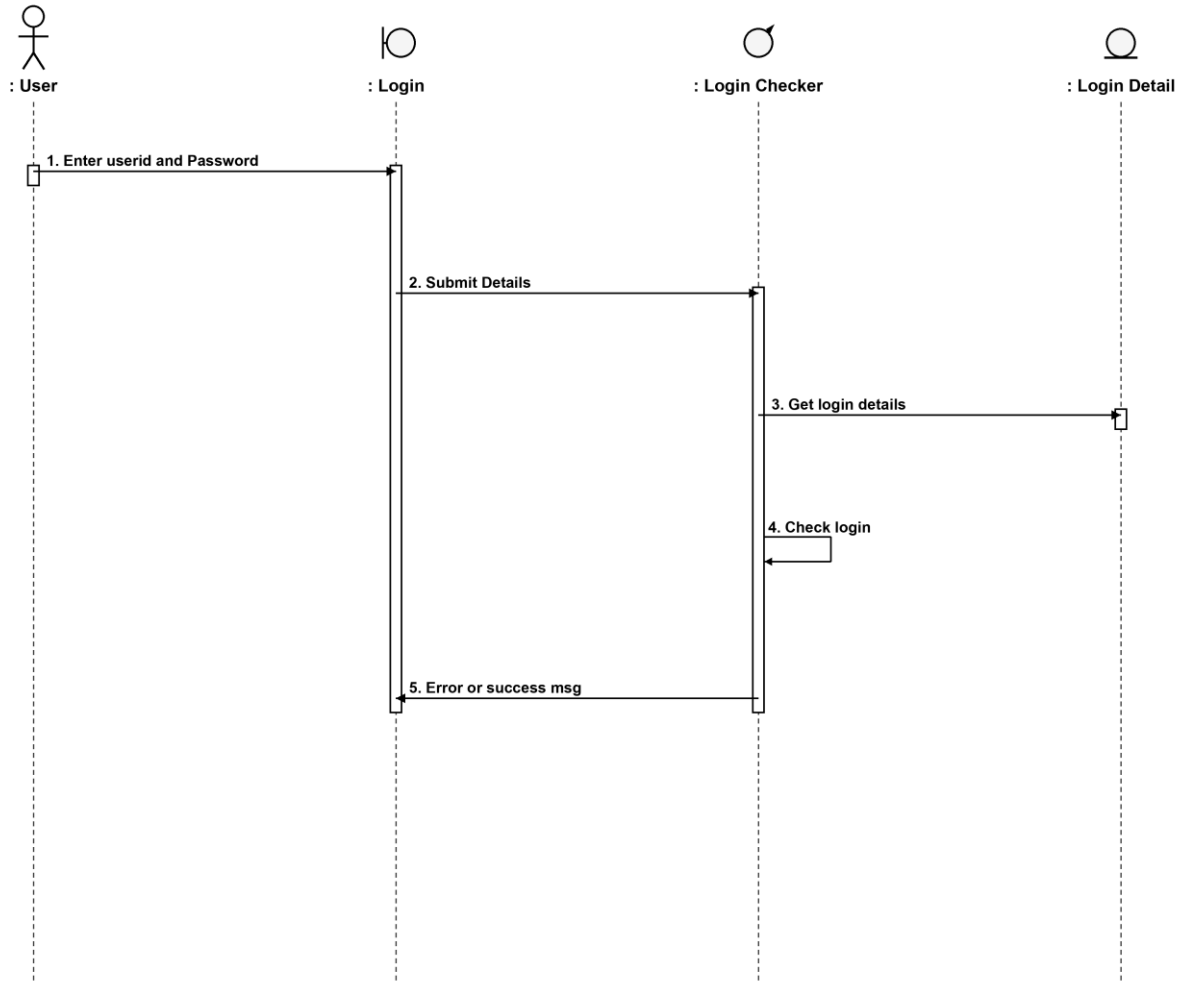
Object-oriented static class structure detailing entities, attributes, methods, and multiplicities.



Class diagram of entity classes

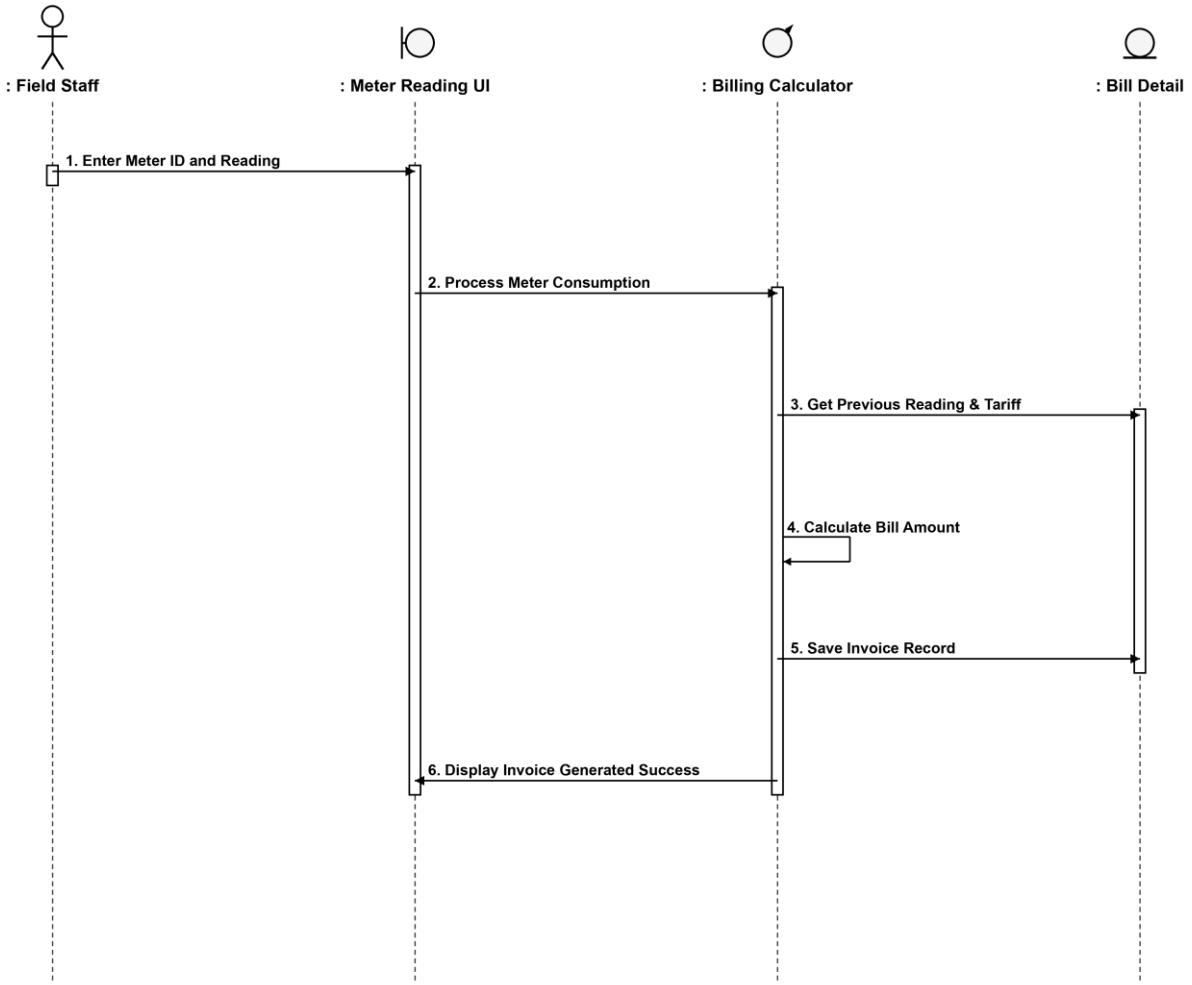
3. Sequence Diagrams

3.1 User Authentication & Session Creation



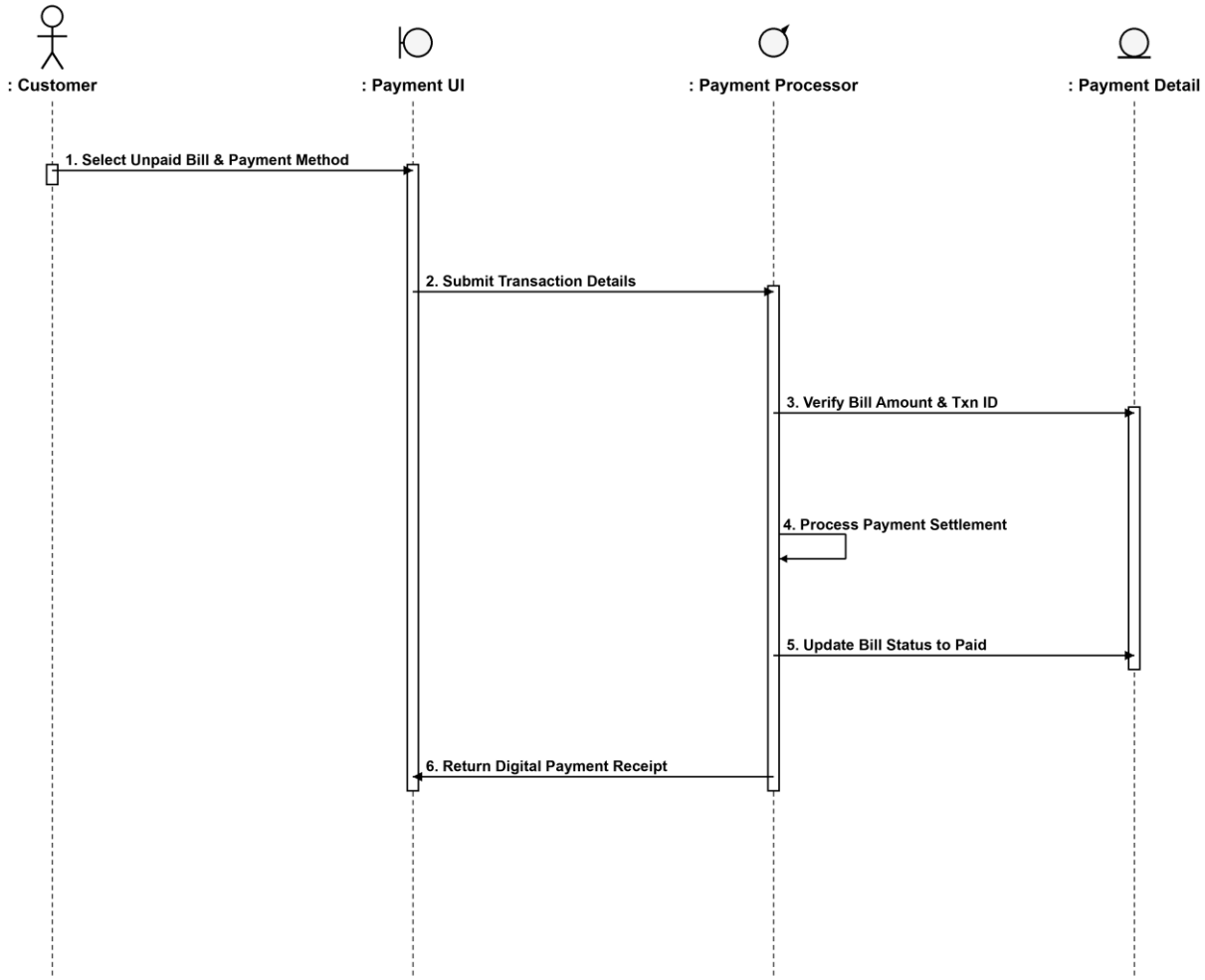
Sequence diagram—Login & User Authentication

3.2 Meter Reading & Automated Billing Process



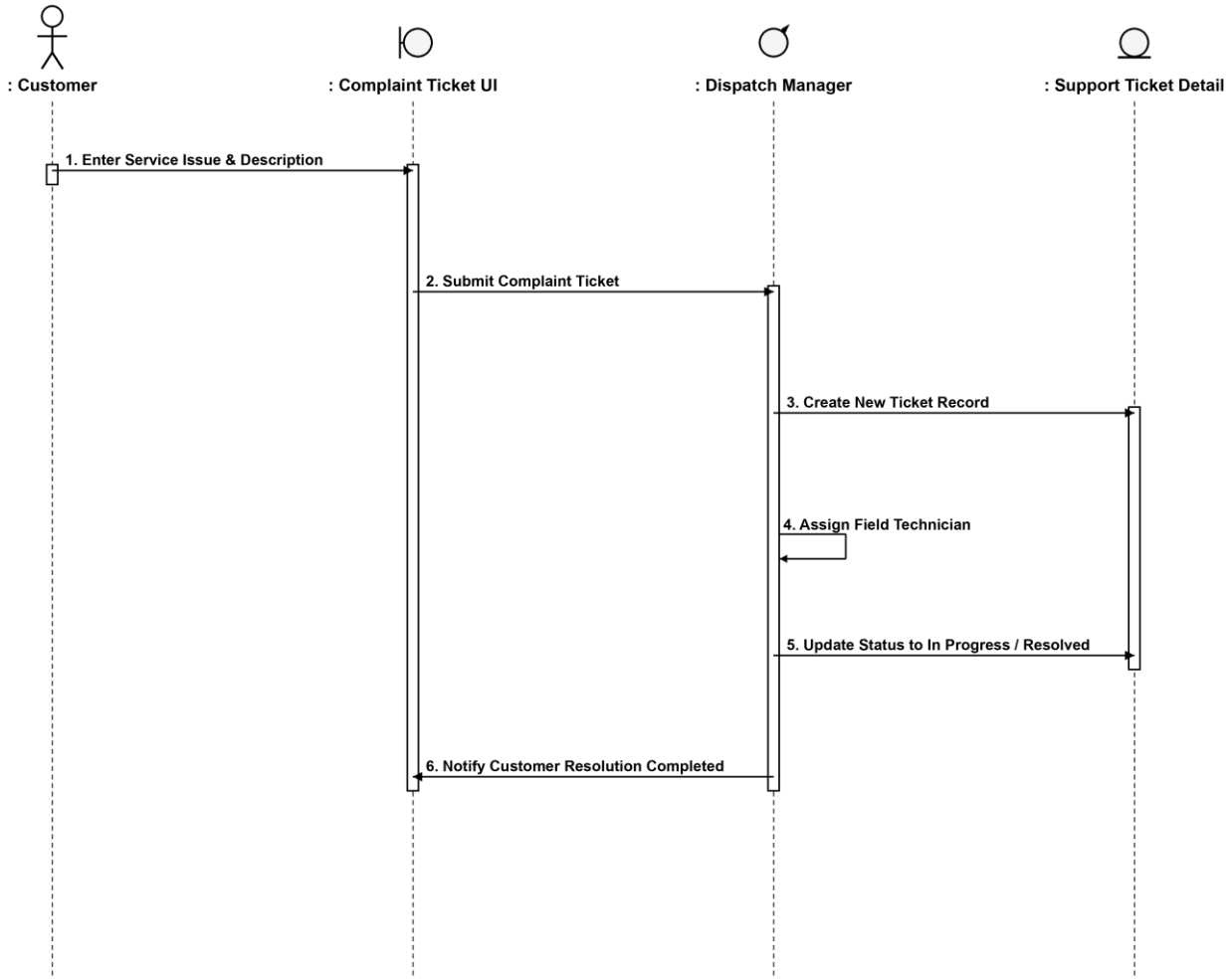
Sequence diagram—Meter Reading & Automated Billing

3.3 Online Bill Payment & Transaction Settlement



Sequence diagram—Online Payment Settlement

3.4 Support Ticket & Field Maintenance Dispatch



Sequence diagram—Support Ticket & Maintenance Dispatch

4. Software Design Diagrams Inventory & Summary

Diagram Name	UML Type	Key Lifelines / Entities	System Design Purpose
Use Case Diagram	Behavioral	System Boundary, 4 Actors, 10 Use Cases	Defines complete system functionality and user access permissions.
Class Diagram	Structural	8 Core Classes (User, Customer, Meter, Bill...)	Models database entities, relationships, attributes, and methods.
Sequence Diagram 1	Interaction	User, LoginUI, AuthController, UserDB	Triggers user credential validation and session token generation.
Sequence Diagram 2	Interaction	Staff, BillingUI, BillingEngine, BillingDB	Executes meter reading delta calculation and invoice creation.
Sequence Diagram 3	Interaction	Customer, PaymentUI, PaymentGateway, UtilisDB	Processes digital bill settlement and transaction status updates.
Sequence Diagram 4	Interaction	Customer, ComplaintUI, Dispatcher, FieldStaff	Handles customer support ticketing and field technician assignment.